Capacity Lab

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1 Objective of the exercise

Estimate how a strategy's net performance changes as assets under management (AUM) grow, and identify the AUM at which its net information ratio (IR) reaches a chosen minimum. The exercise supports capacity planning: it explains the assumptions behind an indicated capacity, rather than treating capacity as a single fixed property of the strategy.

The main output is a **10th–90th percentile interval for AUM at the selected minimum net IR**, obtained from simulated trading books. The dashboard also reports net IR, annual net alpha, the central capacity estimate and the zero-IR threshold. IR is net alpha divided by tracking error. Use the same return horizon and risk conventions when comparing results.

Current AUM is the strategy anchor. The editable baseline gross alpha is annual alpha at current AUM, before estimated impact costs. Current AUM also anchors historical trading observations. USD 0.5bn is only the lower calculation boundary; its net IR is implied by the model, not fixed at 0.60.

2 Two complementary approaches

2.1 Empirical approach

The **Empirical IR curve** combines observed trading and liquidity inputs with three calibrated model pillars: impact costs, alpha decay and transfer-coefficient (TC) decay. It answers: *what happens to performance if orders grow and the chosen execution and implementation assumptions hold?*

Trading observations inform the impact-cost pillar. Alpha-decay and TC points are user estimates that define relative retention shapes; the combined performance level comes from current-AUM gross alpha. The approach makes these assumptions visible and lets users adjust them in Model Calibration.

2.2 Backtest approach

The **Backtest IR curve** fits a decreasing power curve to AUM and IR observations from optimised historical backtests. It provides a separate estimate of capacity based on the strategy's behaviour in those backtests.

Use comparable observations: the same investment period, risk conventions and cost definitions. The fitted curve can be adjusted in the sidebar. Illustrative dashboard observations must be replaced before treating this comparison as historical evidence.

Read the approaches together. Summary compares their current IR and capacity thresholds. Differences can reveal assumptions that need review. The two curves remain separate; the dashboard does not average them or use the backtest fit to calibrate the empirical pillars.

3 Empirical approach: inputs, pillars and calibration

3.1 Inputs

Start in **Inputs & strategy**. Confirm the universe ADV percentiles and annual parent-order count, and check that historical bucket observations correspond to current AUM.

Input	Purpose and units
Current AUM	Gross-alpha and historical order-size anchor; USD billions.
Baseline gross alpha	Annual alpha at current AUM before impact costs; percent.
ADV percentiles	Locally saved USD daily ADV boundaries; interpolate to approximate a 300-stock universe.
Annual parent orders	971 per year; monthly rebalancing.
Historical ADV buckets	Ratios and costs for fitting; counts and notionals calibrate sampling.
Estimated bucket impact	Positive cost in basis points (bp) per traded dollar.
One-way annual turnover	Twice this fraction gives annual traded value / AUM.
Tracking error	Fixed annual active-return risk, entered in percent.
Calibration triples	Three points each for alpha decay and TC.
Minimum net IR	Performance threshold used to indicate capacity.

Alternatively, upload actual parent orders with positive `q` (order notional) and `adv` (average daily traded value) in the same currency. The CSV supplies order sizes and liquidity; estimated bucket costs still supply the impact-curve calibration.

3.2 Three model pillars

3.2.1 Impact costs: larger orders are more expensive to trade

The **ADV bucket sampler** automatically fits a nonnegative power cost curve to the estimated bucket impacts. There are no per-bucket sliders or manual impact-fit controls. Each bucket has equal weight in fitting; aggregate portfolio costs use traded-notional weights.

Calibrate orders. Approximate 300 stock ADVs by log-dollar interpolation of saved percentiles at midpoint ranks. Fix annual bucket counts to history: 801, 113, 33, 18, 5 and 1 (971 total). For historical traded-value share w_b, count n_b, annual one-way turnover fraction T and AUM A in USD billions, the bucket ticket is:

$$q_b(A) = \frac{2TA10^9 w_b}{n_b}, \qquad x_{bi}(A) = q_b(A)/ADV_i.$$

Sample compatible stocks. Draw n_b stocks uniformly with replacement only among names satisfying bucket b's current-AUM order/ADV bounds. Tickets differ across buckets. Each book preserves historical count and notional shares at current AUM and trades exactly $2TA10^9$ annually. This conditions on observed composition, without recovering actual trades.

Reuse each book at every AUM: scale tickets, keep ADV fixed and reclassify orders, including overflow above 100% ADV. Migration bars show mean simulated frequencies. Daily participation is an execution-time reference, not a bucket cap.

Monthly rebalancing partitions the same annual book without changing its cost. Custom annual counts use largest-remainder allocation with at least one order per populated bucket. Incompatible ADV inputs raise errors. Zero turnover gives zero tickets and drag; historical bucket matching then does not apply.

Annual drag is notional-weighted mean impact times twice one-way turnover. CSV mode bootstraps actual order/ADV pairs. Nonlinear-impact research [3] motivates the fit; extrapolation remains an assumption.

3.2.2 Alpha decay: execution congestion reduces performance

Users specify three **(AUM, percentage of trades taking more than one day, net IR)** points. The dashboard fits two linked power curves: AUM to multi-day trade percentage, then multi-day trade percentage to IR. Their curvatures are inferred automatically.

The starting point is USD 0.5bn with zero multi-day trades. The default assumptions are:

AUM (USD bn)	Trades taking > 1 day	Net IR
0.50	0.00%	0.60
3.00	2.00%	0.57
6.00	10.00%	0.50

These points isolate alpha decay while other effects are held fixed. Their IR ratios define relative retention; they do not fix the absolute net-IR level of the combined model. Multi-day percentage counts parent orders equally, regardless of notional or forecast strength. It summarises congestion without distinguishing, for example, two-day and ten-day orders.

3.2.3 Transfer coefficient: implementation efficiency declines

Users specify three **(TC, AUM, annual alpha)** points. The dashboard fits power curves from AUM to TC, then TC to annual alpha. In this exercise, alpha means annual net alpha with other effects fixed.

The default assumptions are:

TC	AUM (USD bn)	Annual alpha
60.00%	0.50	1.00%
40.00%	3.00	0.60%
25.00%	6.00	0.30%

The combined model uses this alpha relationship as a **relative retention factor**: divide calibrated alpha at each AUM by its initial value. The defaults therefore retain 100.00%, 60.00% and 30.00% of the TC-related alpha at the three AUM points. It does not use the raw TC ratio as the retention factor.

These triples imply nonlinear AUM-to-TC decay and a linear TC-to-alpha link. Other triples can imply different curvature. The initial 1.00% normalises this pillar. The combined model has one absolute anchor: gross alpha at current AUM.

3.3 Calibration workflow

1. Set current AUM, gross alpha and tracking error in Capacity Lab.
2. Confirm ADV, annual order count and turnover; inspect the impact calibration.
3. Edit alpha-decay triples and review relative retention.
4. Edit TC triples and review relative retention.
5. Review both approaches in Summary; save via Inputs & strategy.

3.4 How the pillars combine

Multiply the alpha-decay and TC-alpha retention factors to obtain $R(A)$. Normalise this combined shape at current AUM, so the gross-alpha curve passes exactly through the user-entered current-AUM gross alpha. Deduct full estimated annual impact costs once:

$$\boxed{IR(A) = \frac{G_{current}\, R(A)/R(A_{current}) - C(A)/100}{TE}.}$$

Here $G_{current}$ is annual gross alpha in percent, $C(A)$ is annual impact cost in bp and TE is tracking error in percent. Alpha retention is the isolated alpha-decay IR divided by its initial calibration value; TC retention is calibrated TC alpha divided by its initial value. Current AUM must have positive combined retention.

At current AUM, gross alpha equals $G_{current}$ in every simulation. For example, 0.60% gross less 13.00bp impact gives 0.47% net alpha and IR $0.47/1.30 \approx 0.36$. USD 0.5bn is the starting AUM, with model-implied net IR.

3.5 Reading the results

Empirical IR curve shows net IR and net alpha across AUM. **Summary** contains the capacity results and comparison tables for the empirical and backtest approaches; the simulation histogram is last. The pillar-loss breakdown is in Empirical IR curve.

Capacity is the first AUM from USD 0.5bn where the central curve reaches the selected threshold. A threshold not reached within the displayed range does not imply unlimited capacity. Projections beyond calibration points are labelled extrapolation. The combined domain stops when either retained-alpha factor first reaches zero, or at an earlier pillar-domain limit.

The default is **Monte Carlo**: simulate 2,000 complete IR curves and show their pointwise 10th–90th percentile band. **Deterministic** remains available without intervals. Stock selection (or CSV order resampling) and an independent final-IR Gaussian shock vary; alpha-decay and TC relationships stay fixed.

Capacity is the objective. Solve the first AUM at minimum IR for each simulated curve. Summary reports the 10th, 50th and 90th percentiles of these crossings, their distribution and the share reaching the threshold within the horizon. P10–P90 is the central 80% of conditional capacity outcomes, not the crossing of the pointwise IR band.

If a simulation does not cross before the chosen horizon or model-domain limit, its capacity is recorded as beyond that limit. Keep these outcomes in the percentile calculation; do not drop them or assume they cross at the limit. A percentile beyond the limit is displayed as a lower bound. Tables display at most two decimals; calculations and CSV exports preserve full precision.

A Simulation assumptions and calculation notes

A.1 What Monte Carlo samples

A sample is an alternative annual trading book, not a draw of individual trade alpha. Stock selection and an optional final-IR Gaussian shock vary; alpha-decay and TC calibrations stay fixed. The diagram follows one complete simulation to its capacity crossing.

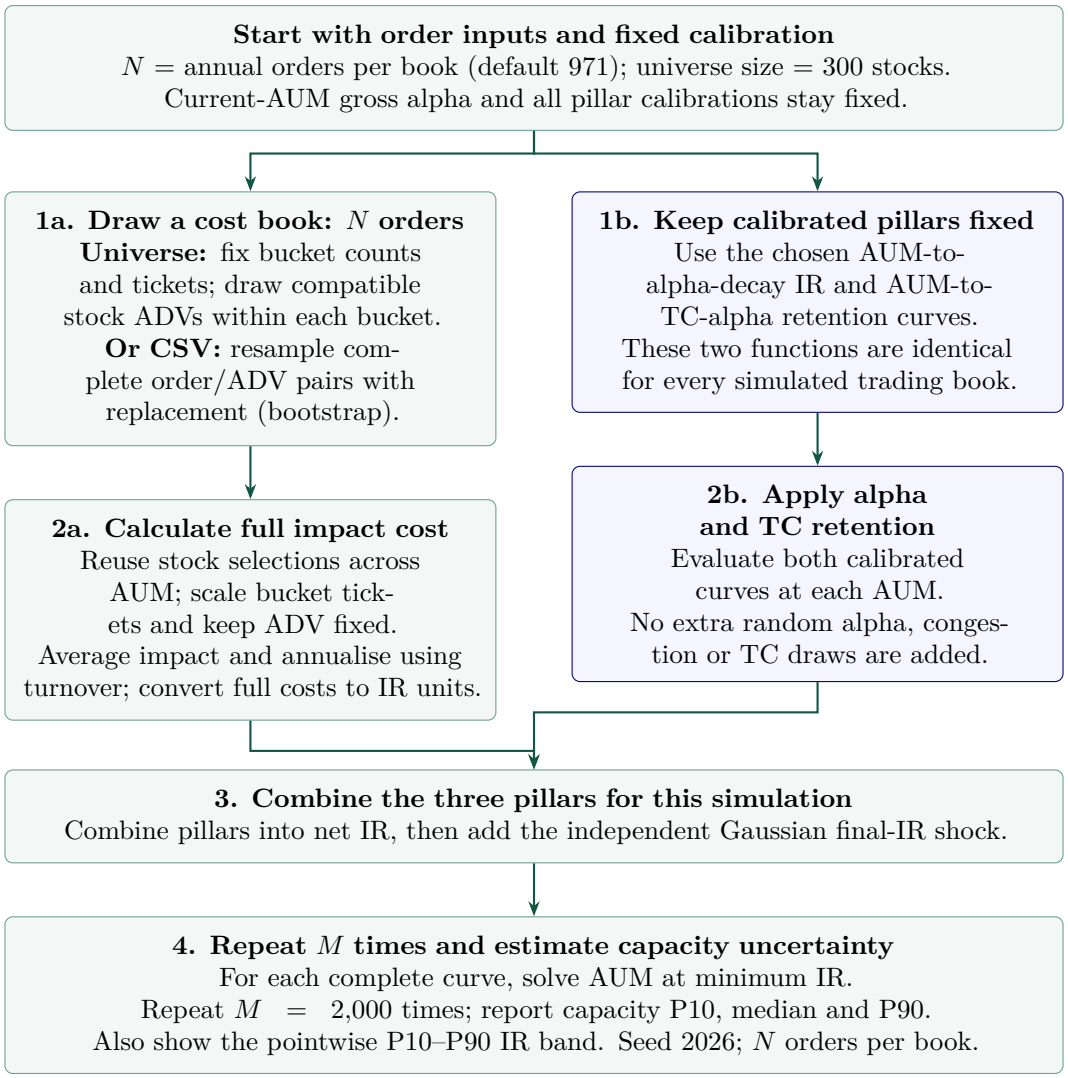


Figure 1. Sampling workflow. Reuse the same sampled book at every AUM. Gross alpha at current AUM is fixed; net IR varies through sampled impact costs. Before the final shock, IR falls as costs rise and retention declines. The Gaussian shock can cause recoveries. The central curve retains zero expected final shock.

What is random? Within each historical bucket, stock counts follow a multinomial distribution with fixed n_b and equal probabilities over its compatible stocks. Baseline bucket counts and notional shares are fixed; participation within buckets and future migration vary. Multinomial sampling is a standard count model [1]; CSV resampling is the empirical bootstrap [2]. The seed makes draws reproducible.

A.2 Backtest uncertainty: wider ranges at higher AUM

Keep the fitted backtest curve $\mu(A) = a - bA^{\gamma}$ as the central estimate. Set the IR standard deviation s_0 at current AUM A_0, and the expansion rate k:

$$\sigma(A) = s_0(A/A_0)^k, \qquad IR_j(A) = \mu(A) + Z_j\sigma(A), \quad Z_j \sim N(0,1).$$

Defaults are $s_0 = 0.03$ IR and $k = 1$: deviation doubles at twice current AUM. Rate zero gives constant width; rate two quadruples deviation at twice current AUM. Zero deviation removes uncertainty. These are user assumptions, not standard errors estimated from saved backtest observations.

Expansion rate	SD at current AUM	SD at $2A_0$	SD at $3A_0$
0 (constant)	0.03	0.03	0.03
1 (linear)	0.03	0.06	0.09
2 (quadratic)	0.03	0.12	0.27

The pointwise P10–P90 band is $\mu(A) \pm 1.2816\sigma(A)$, the central 80% of each Gaussian marginal. Draw one shock per complete path and reuse it across AUM, so path errors are perfectly correlated. This avoids independent point noise; it is an explicit modelling assumption, not an estimated correlation structure.

For every simulated path, solve the first crossing of minimum IR, starting at USD 0.5bn. Large expansion can make optimistic paths turn upward: the search checks the path minimum as well as its endpoint. Paths already below the threshold start at USD 0.5bn; unreached paths remain right-censored beyond the backtest horizon. Retain them when calculating capacity P10, median and P90.

Use the shared sample count and seed with a separate random stream from liquidity sampling. The backtest search uses the selected chart horizon independently of the empirical retention boundary. Summary compares both capacity ranges and histograms. Deterministic mode hides result bands and capacity intervals. Repeated backtests or a return-series block bootstrap could later supply evidence for these uncertainty inputs.

A.3 Final empirical IR uncertainty and capacity quantiles

After applying all three pillars, add an independent Gaussian term:

$$IR_j(A) = IR_{pillars,j}(A) + Z_j s_E(A/A_0)^{k_E}, \quad Z_j \sim N(0,1).$$

Here A_0 is current AUM; s_E is IR deviation there and k_E controls expansion. Defaults are 0.03 IR and 1. Set deviation to zero for stock-selection uncertainty alone. Controls are in the empirical sidebar menu and Empirical IR curve tab; they save with the strategy.

Pair each sampled cost book with one independent shock, reused across AUM. The combined distribution includes both sources and need not be Gaussian. The central curve and its current-AUM anchor remain unchanged. Pointwise bands use combined simulated IR quantiles.

Capacity P10, median and P90 are quantiles of full-path first crossings, not intersections of the pointwise band with the threshold. Search a shared 1,025-point AUM grid plus calibration knots, refine intervening local minima and bisect the earliest crossing. This numerical search handles paths which cross then recover; unreached paths remain censored. The Gaussian term is assumed residual model uncertainty, not evidence-based parameter standard errors.

A.4 Interpretation and limitations

For alpha-decay purposes, orders are exchangeable: the model has no trade-specific alpha forecasts or delay sensitivities. This is a count-based approximation, not an explicit allocation of equal alpha dollars to every trade. Cost aggregation remains notional-weighted; bucket-specific tickets make these weights different from count weights.

Random components are stock selection and the independent final-IR Gaussian term. Alpha-decay, congestion percentages, TC, tracking error and fitted cost parameters stay fixed across simulations. Difficult-to-execute trades may carry more or less alpha in reality; the ranges exclude this dependence, calibration uncertainty, clustered orders and changes in market liquidity. The universe sampler also assumes conditional uniform stock selection, constant tickets within each bucket and a percentile-derived ADV approximation. Baseline historical bucket counts and value shares are imposed rather than random. The IR band is pointwise; the capacity interval is the distribution of per-path first crossings. Both are conditional simulation ranges, not simultaneous confidence bands or guarantees of future performance. The centre curve uses mean simulated impact and zero expected final shock. A finite simulation mean may differ slightly. There is no forced collapse of the range at USD 0.5bn or current AUM: gross alpha is fixed, while impact costs vary.

The Fundamental Law of Active Management motivates separating forecast skill, independent breadth and implementation efficiency. This dashboard does not estimate skill or independent breadth, and its TC scenario is not an estimated forecast-to-position correlation [5, 6].

A.5 Units and fitted relationships

AUM is in USD billions. Order/ADV is a fraction internally and a percentage in the interface. Annual gross alpha, net alpha and tracking error use percentage points. For annual impact $C(A)$ in bp, cost in IR units is $C(A)/(100\,TE)$. Full costs are deducted from gross alpha. The optional breakdown measures losses relative to the model-implied net IR at USD 0.5bn; it does not deduct baseline costs again.

A.6 Academic foundations and model choices

The references support the sampling methods and the economic motivation for the pillars. Efron describes the bootstrap [2]; Agresti covers binomial and multinomial count models [1]. Almgren and co-authors study nonlinear market impact [3]. Gârleanu and Pedersen analyse trading with decaying return predictors and transaction costs [4]. Grinold and Clarke and co-authors provide the active-management and implementation-efficiency context [5, 6].

What the literature does not establish here. Our fixed bucket counts, bucket-specific tickets, conditional stock selection, interpolated ADV universe, three calibration points, fitted congestion-to-IR curve, TC-alpha retention mapping, fixed alpha and TC curves and multiplicative combination are choices for this dashboard. The cited works do not validate these choices or turn the simulated percentile ranges into full forecast confidence intervals.

References

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